

Data Analysis Report: GlobalMart Sales Performance

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Executive Summary

This report analyzes GlobalMart's sales and profitability performance from 2022 to 2024 to identify key drivers of business performance and areas for improvement. The analysis reveals strong overall revenue generation of approximately \$1.08M and profit of \$326K, with an average profit margin above 30%.

The West region contributes the highest share of both sales and profit, indicating strong performance but also exposing the business to regional concentration risk. Product-level analysis shows that high-margin sub-categories such as Binders, Chairs, and Pens outperform lower-margin categories such as Tables and Paper.

Additionally, the Customer Analysis section—based on order-level insights due to data limitations—highlights variations in sales and profit per order, indicating that not all transactions contribute equally to profitability. Discount analysis further shows that higher discount levels are associated with reduced profit margins.

Based on these findings, the report recommends focusing on high-margin products, reducing dependence on a single region, optimizing discount strategies, and improving order-level efficiency monitoring.

1. Business Understanding

1.1 Business Problem and Objectives

The objective of this project is to evaluate GlobalMart's sales and profitability performance to support strategic decision-making. The focus is to move beyond total revenue and understand which products, regions, and order patterns drive profitability.

Key Business Questions

- Which sub-categories drive the highest sales and profit?
- Which regions contribute most to business performance?
- Is revenue growth translating into sustainable profitability?
- What actions can improve profitability and long-term growth?

1.2 Project Goals and Success Criteria

The project is successful if it delivers:

- Clear identification of high and low performing products
- Insights into regional performance differences
- Data-driven recommendations
- A complete Power BI dashboard with five analytical views

2. Data Understanding

2.1 Data Source

The dataset used is GlobalMart_Sales_Data.xlsx, containing transactional sales records from 2022 to 2024.

2.2 Data Dictionary (Summary)

The dataset includes key fields such as Order ID, Order Date, Ship Date, Product Name, Category, Sub-Category, Region, Sales, Profit, Quantity, and Discount.

Column Name	Description	Data Type	Example Value	Notes / Quality Issues
Order ID	Unique identifier of order	Text	f35d7567	Order ID was too long
Order Date	Transaction Date	Date	12/17/2023	Not consistently typed as Date (needs conversion); has blanks/invalid rows
Ship Date	Date of shipping	Date	11/16/2024	Lack of standadization
Product Name	Name of the item	Text	Chairs according	Lack of standadization
Region	Geographical sales region	Text	WEST	Lack of standardization. Replaced values WEST to West.
State	State within region	Text	New York	Lack of standardization
Category	High-level product grouping	Text	Office Supplies	Lack of standardization

Sub-Category	Detailed product segmentation	Text	Accessories	Lack of standardization
Quantity	The amount of the items	Whole Number	10	Removed Errors
Sales	Revenue generated per transaction	Currency	346.03	Format standardization, removing null values (70 of them)
Profit	Net profit per transaction	Currency	429.65	Format standards
Discount	Discount percentage applied	Decimal	0.04	Format standardization
Order Date	Transaction date	Date	8/23/2024	Consistent format used for time intelligence

2.3 Data Limitations

The dataset contains order-level transactional data and does not include a Customer ID field. As a result, customer-level analysis such as retention and repeat purchasing behavior could not be performed. The analysis therefore focuses on product, regional, and order-level performance.

2.4 Exploratory Data Analysis

Initial exploration revealed: - Strong revenue concentration in the West region - Overall profit margins above 30% - Significant variation in profitability across sub-categories

3. Data Preparation

3.1 Data Cleaning

- Standardized inconsistent naming (e.g., region values)
- Removed duplicates and errors
- Corrected data types for date, currency, and numeric fields

3.2 Feature Engineering

The following measures and columns were created:

- Profit Margin (%) = Profit / Sales
- Sales per Order = Total Sales / Total Orders

- Profit per Order = Total Profit / Total Orders
- Date table for time-based analysis
- Month Name and Month Number for proper sorting
- Discount Band for analyzing discount impact

3.3 Data Transformation

Data types were standardized and all numerical fields prepared for aggregation in Power BI.

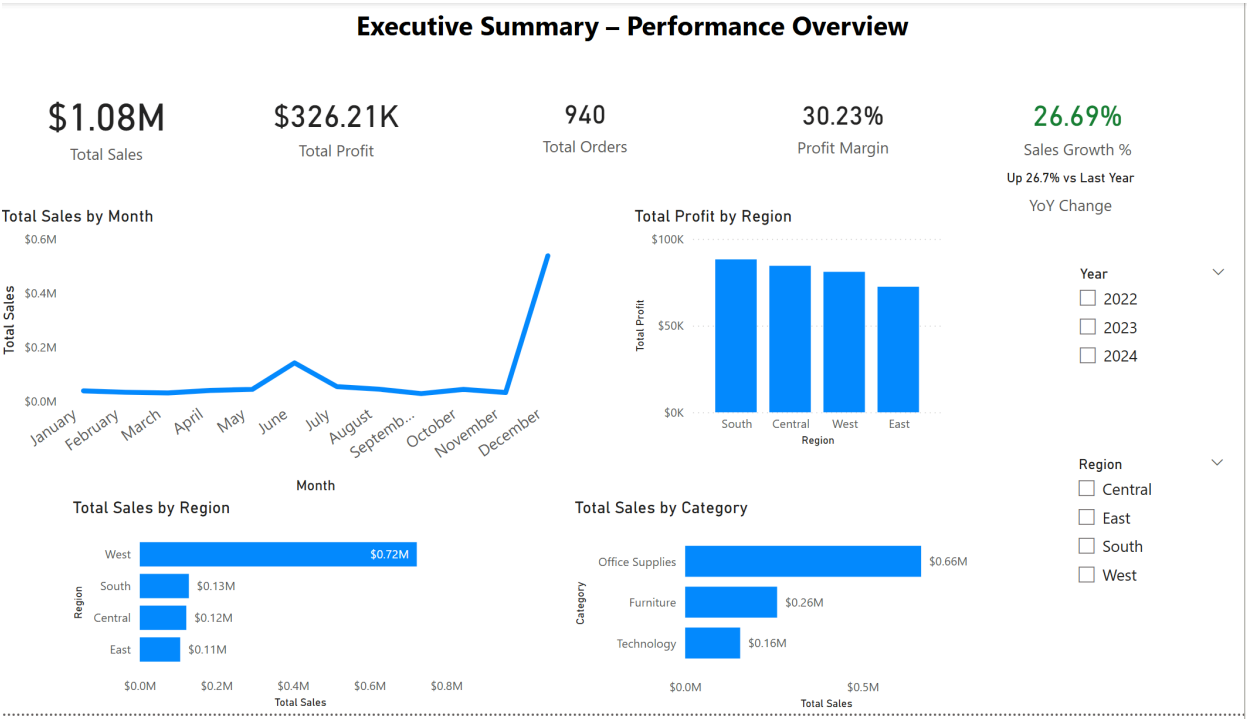
4. Analysis and Findings

Finding 1: Revenue Growth with Strong Profitability

What: The company generated approximately \$1.08M in revenue and \$326K in profit.

So What: This indicates strong financial performance and efficient operations.

Supporting Evidence:

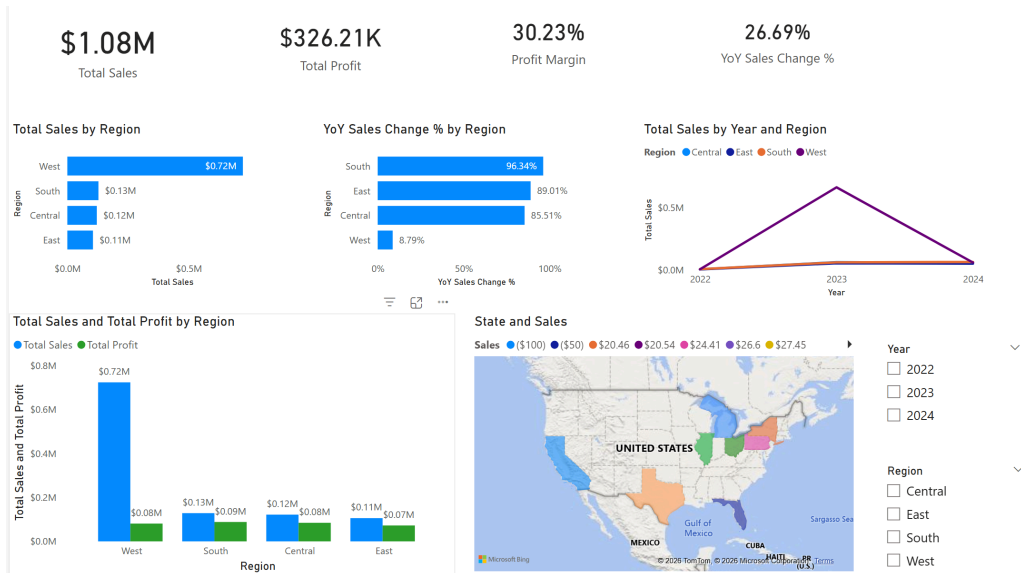


Finding 2: Regional Concentration Risk

What: The West region contributes the highest share of revenue and profit.

So What: Overdependence on one region creates business risk if performance declines.

Supporting Evidence:

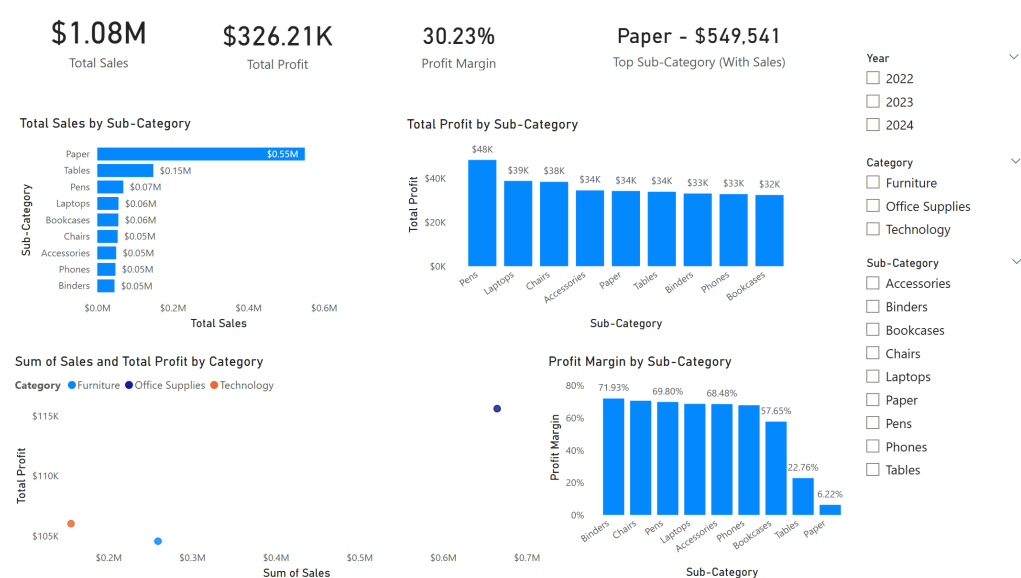


Finding 3: Sub-Category Profitability Variability

What: Profit margins differ significantly across sub-categories.

So What: High sales do not always translate to high profit.

Supporting Evidence:

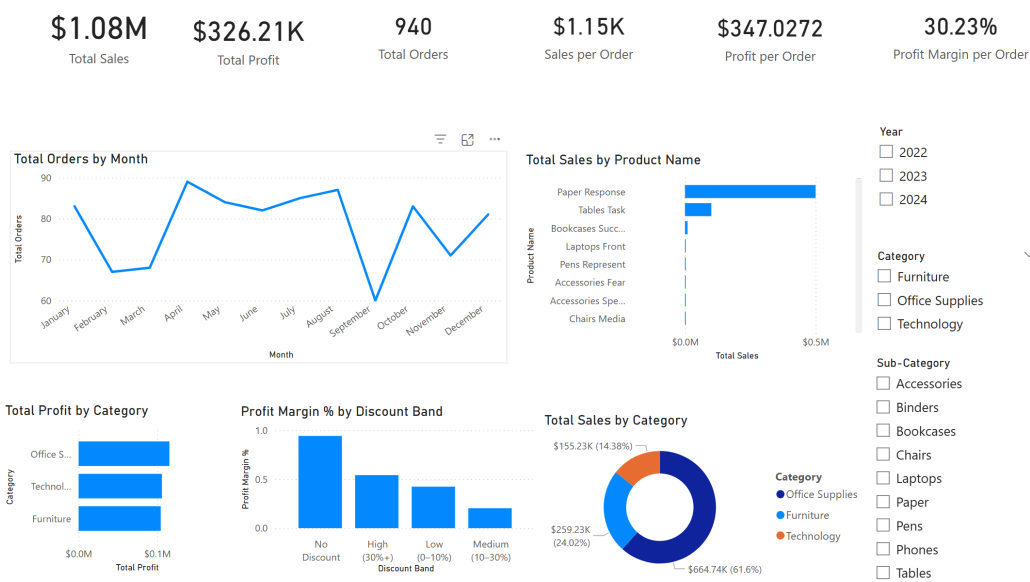


Finding 4: Order Efficiency Differences

What: Sales per Order and Profit per Order vary significantly.

So What: Not all orders contribute equally, highlighting inefficiencies.

Supporting Evidence.

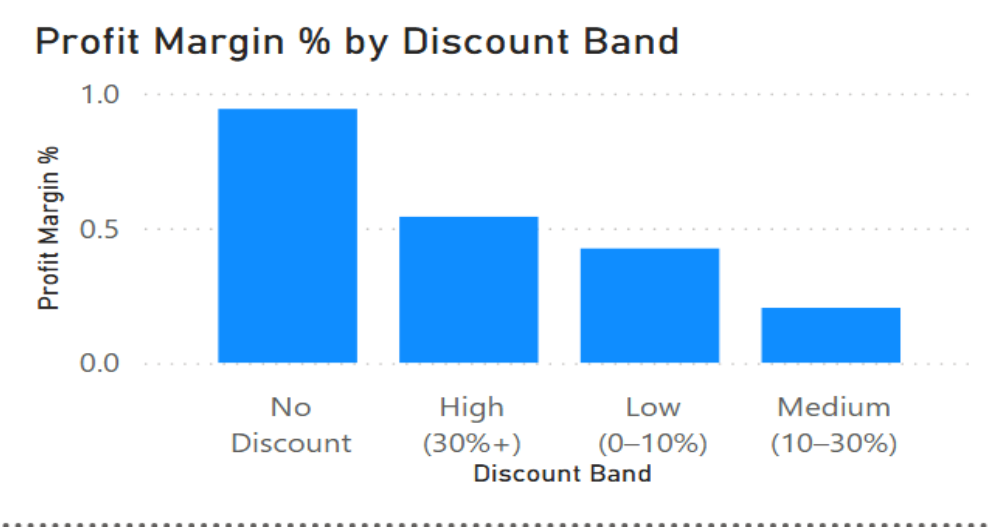


Finding 5: Discount Impact on Profitability

What: Profit margins vary across discount levels.

So What: Discounting can reduce profitability if not controlled.

Supporting Evidence:



5. Dashboard Overview

The Power BI dashboard consists of five key pages:

1. Executive Summary – KPIs overview
2. Sales Overview – trends over time
3. Regional Performance – sales and profit by region
4. Product Analysis – sub-category performance
5. Customer Analysis – insights derived from order-level behavior and discount impact

6. Evaluation of Results

The analysis successfully identified key drivers of sales and profitability. It highlighted regional dependence, product-level inefficiencies, and order-level performance differences.

Limitations

- No customer-level data
- No cost breakdown (e.g., marketing or supply chain)

7. Recommendations and Next Steps

1. Prioritize high-margin sub-categories to improve profit, not just sales

The analysis shows that profit margins vary widely across sub-categories. On the chart in the report, Binders (71.93%), Chairs (69.80%), Pens (68.48%), Accessories, and Phones are among the strongest margin performers, while Tables (22.76%) and Paper (6.22%) are much weaker. This means not all sales contribute equally to profitability. GlobalMart should therefore allocate more inventory, promotions, and sales focus toward the sub-categories with consistently stronger margins, while being more selective with low-margin items. This will help the business shift from revenue growth alone to profitable revenue growth. Based on this margin spread, the business can reasonably improve overall profitability by increasing the sales mix of high-margin products and reducing over-reliance on low-margin lines.

Actionable next steps

- Increase marketing emphasis on high-margin sub-categories such as Binders, Chairs, Pens, Accessories, and Phones.
- Review stock planning to avoid understocking profitable sub-categories.
- Set profitability targets by sub-category, not just sales targets.

2. Reduce overdependence on the West region through regional growth balancing

Your report shows that the West region generates about \$0.72M in sales, which is far ahead of South (\$0.13M), Central (\$0.12M), and East (\$0.11M). Profit also follows the same pattern,

with West contributing the highest share. While this is positive in the short term, it creates regional concentration risk because the business is heavily dependent on one market. If performance drops in the West region, the company's total revenue and profit would be significantly affected. Instead of only maintaining West, GlobalMart should invest in growing the underperforming regions so that business performance becomes more balanced and resilient.

Actionable next steps

- Run targeted sales campaigns in South, Central, and East where current revenue is much lower than West.
- Compare product mix by region to identify which profitable categories are underpenetrated outside West.
- Set region-specific growth targets to reduce overconcentration.

3. Review discounts on low-margin products to protect profit

In the EDA section, you noted that some transactions had reduced profitability because of higher discount application. At the same time, some sub-categories already have very weak profit margins, especially Paper and Tables. This suggests that discounting may be eroding profit further in products that are already margin-sensitive. A blanket discount strategy is risky because it can grow sales volume without improving business performance. GlobalMart should therefore review its discount policy by product type and limit aggressive discounting on weak-margin items unless there is a strong strategic reason such as stock clearance or market entry.

Actionable next steps

- Create discount thresholds by sub-category based on historical profitability.
- Flag products with low margins and high discounts for management review.
- Use discounts more selectively on products that can absorb them without hurting profit.

4. Track growth quality, not just revenue growth

Findings state that the company generated approximately \$1.08M in revenue, achieved a profit margin above 30%, and recorded 26% year-over-year sales growth. That is a strong result, but management should not assume that every increase in sales will remain equally profitable in the future. The next step should be to track whether future growth is coming from strong-margin regions and sub-categories or from discounted, low-margin sales. This shifts the business from descriptive analysis to performance management.

Actionable next steps

- Add KPI tracking for sales growth, profit growth, and profit margin trend together.
- Monitor whether YoY growth is being driven by healthy product segments.
- Create monthly exception reporting for fast-growing but low-margin areas.